



Product Requirements Document (PRD)

Project Title: Library Digital Token System – IIT Guwahati Library

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1. Introduction

The IIT Guwahati Library is one of the most frequented spaces on campus, with hundreds of students visiting daily. Due to bag restrictions, a **manual bag deposit system** is in place, where students hand over their bags at the entrance and receive a **metallic token** corresponding to a storage slot.

While functional for years, this system has become increasingly inefficient and frustrating. Students and guards regularly face issues with **lost tokens, long wait times, mix-ups, and disputes**.

Meanwhile, students already rely heavily on the **OneStop app** for daily campus activities (timetables, hostel logs, fests, lost & found, transactions). Leveraging this digital adoption, integrating the library bag deposit system into OneStop can deliver a **seamless, token-free experience**.

2. Problem Statement

The IIT Guwahati Library's **manual token system for bag storage** has become increasingly inefficient and stressful for students and staff. Students must carry **small metallic tokens** to claim their bags, but these are often misplaced or forgotten, leading to a steep **₹1000 fine** and unnecessary anxiety. Even when tokens are retained, students frequently forget their numbers, causing delays and confusion during retrieval.

For library guards, the system is equally challenging. Without a digital record, managing deposits and retrievals becomes error-prone, especially during peak hours and shift changes. Disputes over **misplaced tokens or slot assignments** add to operational stress and slow down the process.

During high-traffic periods such as **exams, long queues** form, creating frustration for students and further complicating guards' workflows. Occasional mix-ups and near-misplacements of valuable items have also eroded trust in the system's

reliability. Overall, the current manual process is **outdated**, inconvenient, and unable to meet the expectations of a digitally connected student body.

3. User Research & Validation

Links :

- Library Token Digitalization Survey Link : [LINK](#)
- Student Interviews : [DRIVE FOLDER](#)

To validate our initial hypotheses, a mixed-method research approach was employed, combining qualitative user interviews with a quantitative campus-wide survey. This approach allowed us to understand not only *what* the problems are but *why* they are such significant pain points for students.

Research Method

- **Interviews:** We spoke with undergrad and postgrad students to hear about their experiences with the bag deposit system.
- **Survey:** A Microsoft Form got 140+ responses, giving us solid data on student opinions.

Survey Results

- **Unhappy Students:** 82% rated the system **Poor** or **Very Poor**.
- **Costly Fines:** 35% have lost tokens and paid the **₹1000 fine**.
- **Forgetfulness:** 65% often forget their token at different places, causing delays.
- **Long Queues:** 78% faced big delays during exam time.
- **Want Digital:** 92% prefer a QR-based system.

What Students Said

- **Stress:** The ₹1000 fine creates constant worry.

- **Slow Process:** Manual handling of tokens causes delays, especially at peak hours.
 - **Security Issues:** Students don't fully trust the system, with reports of bag mix-ups.
 - **Token Hassles:** Tokens are easy to lose and even scratch phones.
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4. PAIN POINTS

- **Risk of Token Loss** →
 1. Students face a **heavy fine of ₹1000** if they lose the token.
 2. Students often spend a long time anxiously looking for the misplaced token.
- **Inconvenience of Carrying Tokens** → Students must carry the token in their pockets or hands, which is uncomfortable. Tokens are **small and easy to misplace**.
- **Long waiting periods** → Students have to wait for long time periods to get their bags back due to the slow manual system, which is frustrating especially during exam time.
- **Forgetting the token in Hostel** → Students tend to forget their token in the hostel when they go for lunch or something and have to go all the way back to the hostel to get their token back.
- **Tokens get exchanged** → Many times the tokens of students get exchanged at the deposit counter itself, leading to further inconvenience.
- **Tokens left at the counter with the guard** → Sometimes students forget to collect their token from the guards after depositing their bag.
- **Bags getting swapped** → Sometimes guards mistakenly give the wrong token and thus it gets difficult to retrieve the bag. Moreover, lookalike bags sometimes do get swapped.

5. Users & Stakeholders

Primary Users – Students

- **Needs:** Quick, reliable, secure bag storage without fines.
- **Pain Points:** Token anxiety, inconvenience, delays.
- **Definition of Success:** Seamless digital token system that reduces mental and physical effort.

Secondary Users – Security Staff

- **Needs:** Simplified workflow with reduced disputes.
- **Pain Points:** Confusion when students forget tokens, delays during peak hours, manual errors.
- **Definition of Success:** Automated system that reduces human error and speeds up retrieval.

Key Stakeholders

- **Library Administration:** Interested in operational efficiency, security, and reduced disputes. Concerned about cost and integration.
- **Students' Web Committee (Technical Team):** Responsible for implementation. Interested in scalability and reliability. Concerned about maintenance and security.
- **IT Infrastructure Team:** Ensures backend integration with OneStop and campus systems.
- **Student Body Representatives:** Advocacy and adoption feedback loop.

6. Current Landscape

Existing Process

1. Student deposits bag → guard allocates slot.
2. Guard issues metallic token (numbered).
3. Student retrieves bag by returning token.
4. Guard cross-verifies and hands back the bag.

Summary: Current system is outdated, stressful, and inefficient → strong mandate for digital replacement.

7. Goals & Objectives

Goals:

1. **Digitize Bag Storage** – Eliminate dependency on physical tokens.
2. **Enhance User Convenience** – Provide a faster, hassle-free deposit/retrieval process.
3. **Improve Operational Efficiency** – Streamline guard workflow and reduce errors.

4. **Leverage Existing Infrastructure** – Integrate within IITG OneStop app.

Objectives:

- Eliminate fines/disputes caused by lost tokens.
 - Reduce average bag retrieval time by >50%.
 - Achieve high adoption (>80% students within 3 months).
 - Improve guard productivity and reduce manual effort.
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8. Proposed Solution: Mobile App–First Digital Storage

The proposed solution is a **mobile app (OneStop)–driven bag storage and retrieval system** that replaces the physical token process with a **secure, digital-first workflow**. Instead of handling tokens, students will rely on their smartphones, while guards will manage deposits through a simple scanning and dashboard interface.

User Flow

Bag Deposit:

1. Student selects “Deposit Bag” in IITG app.
2. A unique digital slot number + QR code is generated.
3. Guard scans the student’s QR code → assigns slot.
4. Student receives a confirmation notification with slot number stored in app.

Bag Retrieval:

1. Student selects “Retrieve Bag” in app.
2. A digital retrieval QR/code is displayed.
3. Guard verifies and hands over bag.
4. Status is updated as “Retrieved” in system logs.

Features

Students
Digital slot storage, no risk of token loss.
Push/SMS notifications for deposit confirmation.
Reminder alerts during library closing hours
Option to regenerate retrieval ID in case of app logout/forgetfulness.
Guards
Real-time dashboard of all deposited bags.
QR scanner for quick verification.
Easy slot lookup by student ID.
Seamless shift handover with persistent records.

Administration
Analytics dashboard for usage, peak hours, and efficiency.
Centralized audit logs for accountability.

Advantages

- Eliminates fines, disputes, and frustration.
 - Saves students' time with quick, digital retrieval.
 - Provides guards a stress-free, error-proof workflow.
 - Builds trust and accountability via digital records.
 - Supports IITG's image as a digitally forward campus.
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9. Technical Requirements

- Integration: With existing IITG app (or a modular standalone).
- Authentication: Secure login via institute credentials.
- Database: Centralized mapping of student ID ↔ slot ↔ status.
- Security: Time-limited QR codes to prevent misuse.
- Scalability: Must handle 10,000+ transactions monthly.

10. Success Metrics

- 100% elimination of token-loss fines.
 - >50% reduction in average retrieval time.
 - >80% adoption rate within 3 months of rollout.
 - Guard satisfaction score >4/5 on usability survey.
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